

PROJECT REVIEW – 1

Student-Exam-Score-Predictor

Machine-learning based prediction of expected examination scores
from academic and behavioral factors

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Outline of Project Work

01 Introduction & Problem Statement

02 Research Objectives

03 Literature Survey

04 Problem Gap Identification

05 Innovation, Creativity & Novelty

06 Feasibility, Challenges & Project Plan

07 Conclusion & Future Work

08 References & Publications Details

Review focus: problem → evidence → gap → proposed solution → feasibility → plan

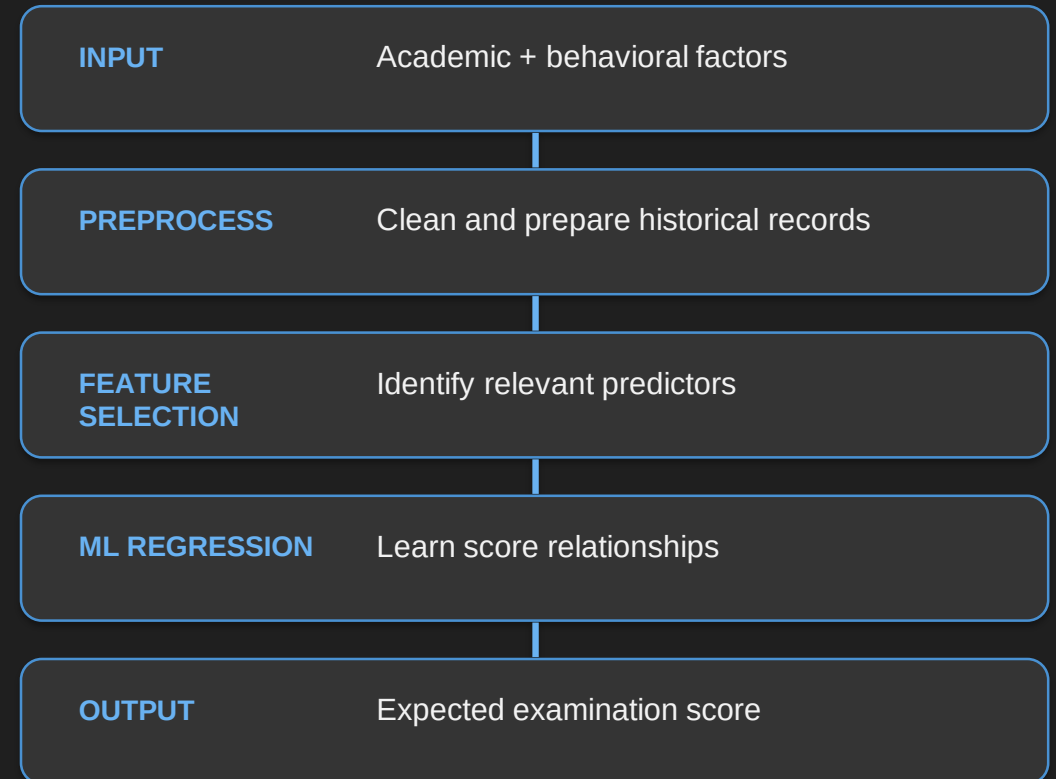
Introduction & Problem Statement

WHY THIS PROBLEM MATTERS

Student performance is multi-factorial

- Study hours and attendance
- Previous examination marks
- Assignment performance
- Participation in academic activities
- Sleep patterns and other learning-related attributes

PROPOSED IDEA



Goal: make academic performance assessment more informative and proactive.

Research Objectives

01

Collect

Gather academic and behavioral factors relevant to examination performance.

02

Preprocess

Prepare historical student records for reliable modelling.

03

Select Features

Identify factors that are most relevant to academic performance.

04

Train Model

Apply a suitable machine-learning regression algorithm.

05

Predict

Estimate an expected examination score for new student inputs.

06

Support

Help students and institutions identify areas needing academic support.

END-TO-END FLOW

Student factors

Preprocessing

Feature selection

Regression

Predicted score

Literature Survey

Selected studies relevant to educational data mining and student-performance prediction

Authors / Year	Title / Focus	Methodology	Key inference	Limitation / relevance
Dabhade et al. (2021)	Educational data mining for predicting students' academic performance	Preprocessing + factor analysis; compares regression models	Academic, personal and behavioral attributes can support prediction.	Institution-specific dataset; motivates broader project features.
Yağcı (2022)	EDM: prediction of students' academic performance	RF, SVM, LR, NB, kNN and NN for final-exam prediction	Midterm grades, faculty and department were used as predictors.	Limited input features; focuses on prediction/classification settings.
Tomasevic et al. (2020)	Supervised data-mining techniques for student exam performance	Comparison of supervised ML techniques and input datasets	Model performance varies with algorithm and input data.	No single universal model; supports project-specific comparison.
Yuan et al. (2026)	Systematic review of student-success prediction	PRISMA review of studies from 2015–2024	Academic/personal factors and ML methods are repeatedly used.	Highlights need to connect prediction with practical intervention.

Literature basis used in the current review; references are listed on Slide 9.

Problem Gap Identification

What the reviewed literature leaves open for this project

Limited feature coverage

Some studies use a narrower set of academic or demographic variables.

Classification focus

Several approaches focus on pass/fail or risk categories rather than a continuous expected score.

Context dependence

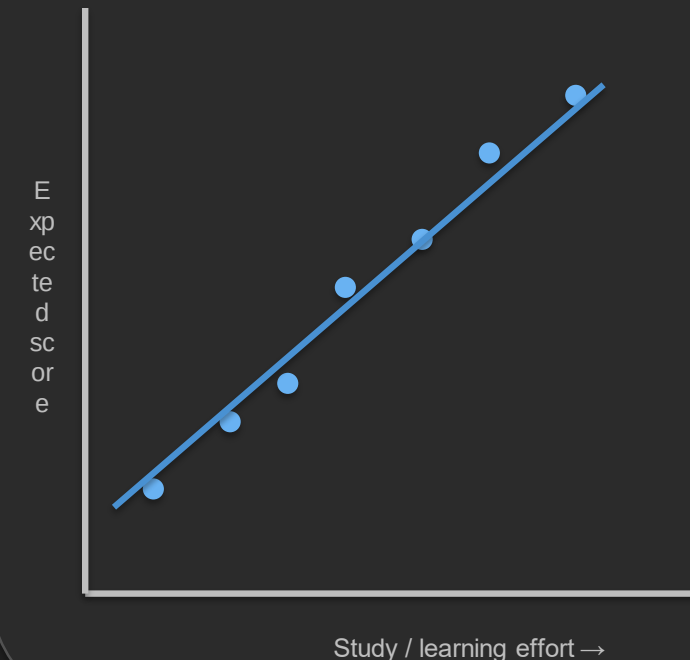
Model performance depends on the selected features, dataset and institution.

From prediction to support

A useful system should turn the prediction into understandable academic guidance.

Conceptual relationship

Illustrative only — not project results



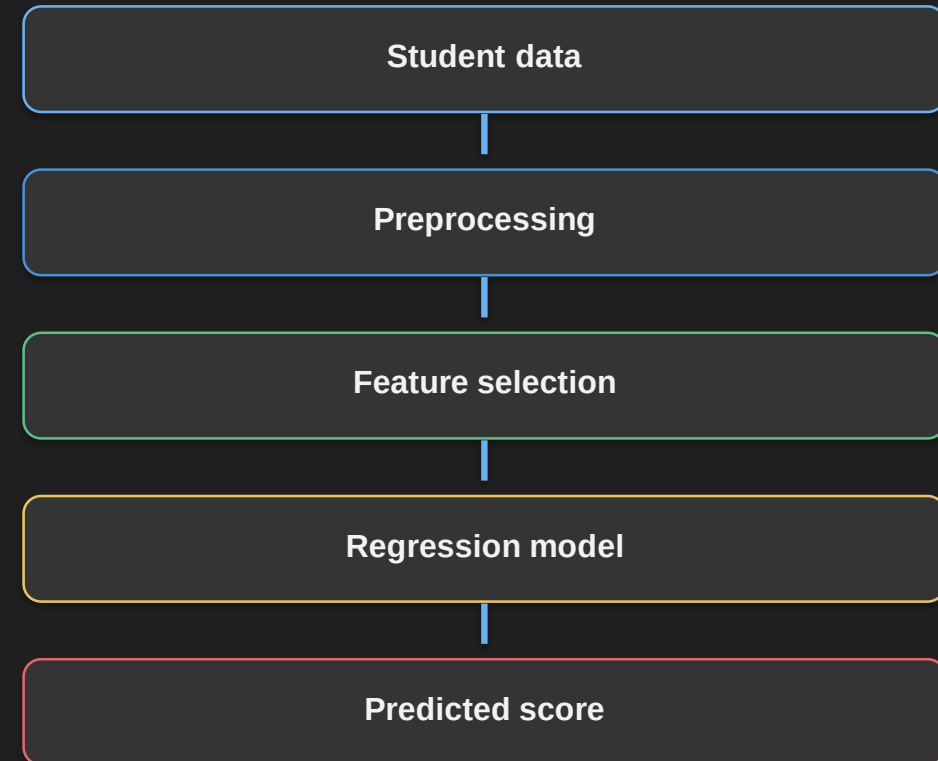
Project response: combine multiple learning-related attributes and predict a continuous expected score.

Innovation, Creativity & Novelty

Proposed contribution

- Combines academic and behavioral factors in one prediction workflow.
- Uses preprocessing and feature selection before model training.
- Targets a continuous expected examination score.
- Produces an output that can support early academic planning.
- Can be extended with explainable AI and a student/teacher dashboard.

System concept



Future: explainability • dashboards • continuous model updating • institution-specific validation

Feasibility and Project Plan

Feasibility

Technical	Python, pandas, scikit-learn and standard ML regression methods are sufficient.
Data	Historical student records and exam scores can be used after preprocessing and privacy protection.
Operational	Results are intended to be simple for students and academic staff to interpret.
Challenges	Missing values, inconsistent records, feature selection, overfitting and generalization.

PROJECT TIMELINE

1	Data collection	Weeks 1–2
2	Data preprocessing	Weeks 3–4
3	Feature selection	Week 5
4	Model training & comparison	Weeks 6–7
5	Prediction module	Week 8
6	Testing & validation	Weeks 9–10
7	Documentation & presentation	Weeks 11–12

Planned outcome: validated score-prediction prototype + documented findings

Conclusion, Future Work & References

Conclusion

- Predict expected examination scores from student-related factors.
- Use data preprocessing, feature selection and regression.
- Support earlier academic assistance and planning.

Future Work

- Explainable predictions
- Student/teacher dashboard
- Continuous model updating
- Institution-specific validation

References

- [1] S. Karthikeya et al., "Student-Exam-Score-Predictor," Project Abstract Submission, A.Y. 2026–2027.
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- [3] M. Yağcı, "Educational data mining: prediction of students' academic performance using machine learning algorithms," Smart Learning Environments, 2022. DOI: 10.1186/s40561-022-00192-z.
- [4] N. Tomasevic, N. Gvozdenovic, S. Vranes, "An overview and comparison of supervised data mining techniques for student exam performance prediction," Computers & Education, 2020. DOI: 10.1016/j.compedu.2019.103676.
- [5] L. Yuan et al., "Predicting student success: a state-of-the-art systematic review of models and factors," Journal of Applied Research in Higher Education, 2026. DOI: 10.1108/JARHE-08-2025-0674.
- [6] M. N. Gul et al., "Data driven decisions in education using a comprehensive machine learning framework for student performance prediction," Discover Computing, 2025. DOI: 10.1007/s10791-025-09585-3.

Publications / paper details: literature sources listed above.